

ASSIGNMENT 7

LFD is the class textbook

1. (500) Classifying Handwritten Digits: 1 vs. 5

Implement the following classification algorithms for non-separable data:

- (i) Linear Regression for classification followed by pocket for improvement.
- (ii) Logistic regression for classification using gradient descent.
- (iii) Logistic regression for classification using stochastic gradient descent.
- (iv) Linear Programming for classification (Graduate, 6xxx-level, only).

Use each method to find a separator *using the training data* and your 2 features from a previous assignment. The target is +1 if the example is a 1 and -1 for a 5.

- (a) For each method, plot the separator with the training and test data (separate plots).
- (b) For each method, compute E_{in} on the training data and E_{test} on the test data.
- (c) Pick the method with the minimum E_{in} and bound the true out-of-sample error using E_{in} and a tolerance $\delta = 0.05$.
- (d) Pick the method with the minimum E_{test} and bound the true out-of-sample error using E_{test} and a tolerance $\delta = 0.05$.
- (e) Repeat (a)-(d) using a 3rd order polynomial transform.
- (f) As your final deliverable to a customer, would you use the linear model with or without the 3rd order polynomial transform? Explain.

2. (200) Gradient Descent on a “Simple” Function

Consider the function $f(x, y) = x^2 + 2y^2 + 2\sin(2\pi x)\sin(2\pi y)$.

- (a) Implement gradient descent to minimize this function. Let the initial values be $x_0 = 0.1$; $y_0 = 0.1$, let the learning rate be $\eta = 0.01$ and let the number of iterations be 50; Give a plot of the how the function value drops with the number of iterations performed.
Repeat this problem for a learning rate of $\eta = 0.1$. What happened?
- (b) Obtain the “minimum” value and the location of the minimum you get for gradient descent using the same η and number of iterations as in part (a), starting from the following initial points: $(0.1, 0.1)$, $(1, 1)$, $(-0.5, -0.5)$, $(-1, -1)$. A table with the location of the minimum and the minimum values will suffice. You should now appreciate why finding the “true” global minimum of an arbitrary function is a hard problem.

3. (300) Problem 3.16 in LFD